

PROJECT REPORT

IMAGE CLASSIFICATION USING DEEP RESIDUAL LEARNING (RESNET-20) ON CIFAR-10

Research-paper-based implementation and experimental study

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Abstract

This project implements a convolutional neural network for image classification based on the residual learning framework proposed in “Deep Residual Learning for Image Recognition” by Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. CIFAR-10 is used as the target dataset. A ResNet-20 model was implemented with 3×3 convolutions, batch normalization, identity residual shortcuts, global average pooling, and a 10-class classifier. The current prototype was trained for 80 epochs using SGD with momentum and weight decay on Apple MPS. It achieved a best observed test accuracy of 90.92% (9.08% error) at epoch 79. The report documents the implementation, results, comparison with the paper, and the methodological correction required before final evaluation.

1. Introduction

The selected paper addresses the degradation problem in deep plain networks. Rather than directly learning an underlying mapping $H(x)$, residual blocks learn a residual function $F(x)=H(x)-x$ and produce $F(x)+x$ through shortcut connections. The paper presents evidence that this formulation makes deeper networks easier to optimize and can improve accuracy as depth increases.

2. Problem Statement

The objective is to build a research-paper-based image-classification model, evaluate it on CIFAR-10, and then study the effect of network depth by extending the implementation from ResNet-20 to deeper residual networks.

3. Objectives

- Study residual learning and shortcut connections.
- Implement ResNet-20 in PyTorch.
- Train on CIFAR-10 with paper-inspired augmentation.
- Measure loss and accuracy.
- Evaluate precision, recall, F1-score and the confusion matrix.
- Compare the implementation with the published results.
- Extend the experiment to ResNet-32, ResNet-44 and ResNet-56.

4. Dataset

CIFAR-10 contains 50,000 training images and 10,000 test images in 10 classes. The images are 32×32 RGB images. The local dataset was loaded from the CIFAR-10 binary files through torchvision. Training used batch size 128.

Property	Value
Training images	50,000
Test images	10,000
Image size	32×32 RGB
Classes	10
Batch size	128

5. Model Architecture

The implemented ResNet-20 starts with a 3×3 convolution producing 16 channels. Three residual stages follow: 16 channels at 32×32, 32 channels at 16×16, and 64 channels at 8×8. Each stage contains three two-convolution residual blocks. Global average pooling is followed by a 10-class fully connected layer. The model has 269,722 trainable parameters.

Stage	Size	Channels	Blocks
Initial	32×32	16	1 conv
Stage 1	32×32	16	3
Stage 2	16×16	32	3
Stage 3	8×8	64	3
Global pool	1×1	64	—
Classifier	—	10	FC

6. Data Augmentation

Training uses RandomCrop(32, padding=4), RandomHorizontalFlip(), and ToTensor(). Test images use ToTensor() only. This closely follows the paper's CIFAR-10 augmentation description of four-pixel padding, random 32×32 cropping and horizontal flipping.

7. Training Configuration

The current implementation uses SGD with learning rate 0.1, momentum 0.9, weight decay 0.0001, batch size 128, and no dropout. The learning rate is reduced by 10× at epochs 40 and 60. Training runs for 80 epochs on Apple MPS. The paper instead uses 64k iterations with reductions at 32k and 48k iterations and reports training on two GPUs.

Parameter	Current	Paper
Optimizer	SGD	SGD
LR	0.1	0.1
Momentum	0.9	0.9
Weight decay	0.0001	0.0001
Batch	128	128
LR schedule	Epoch 40/60	32k/48k iters
Length	80 epochs	64k iters
Hardware	Apple MPS	Two GPUs

8. Experimental Results

Metric	Result
Best observed test accuracy	90.92% (epoch 79)
Best observed test error	9.08%
Final test accuracy	90.86%
Final test loss	0.3322
Final training accuracy	97.67%
Final training loss	0.0697
Macro F1	0.9093

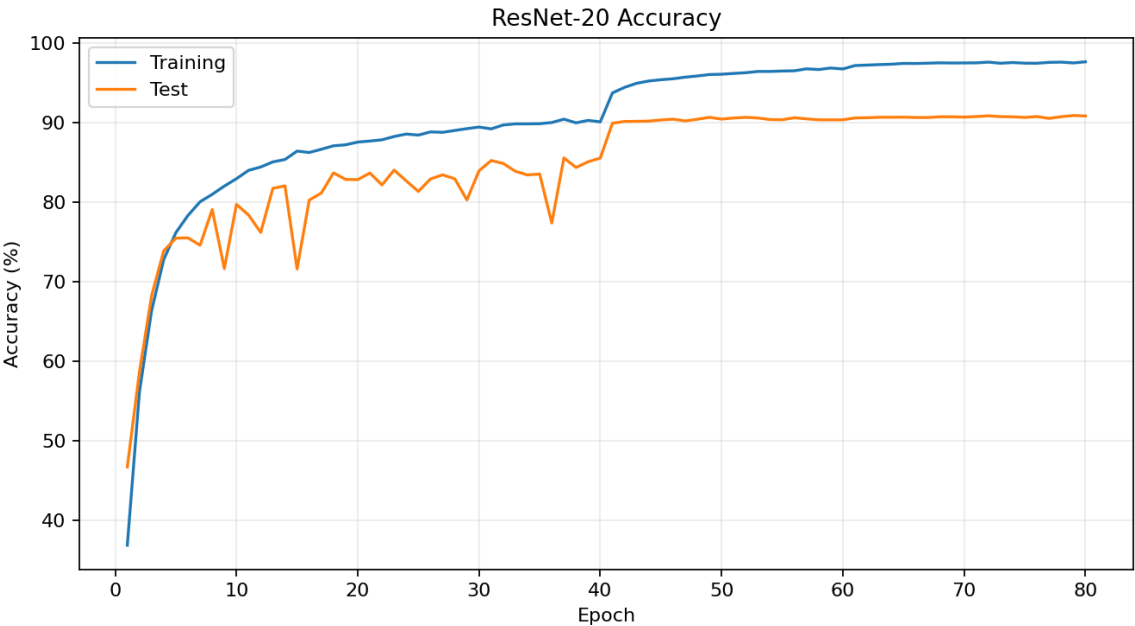


Figure 1. Accuracy curve.

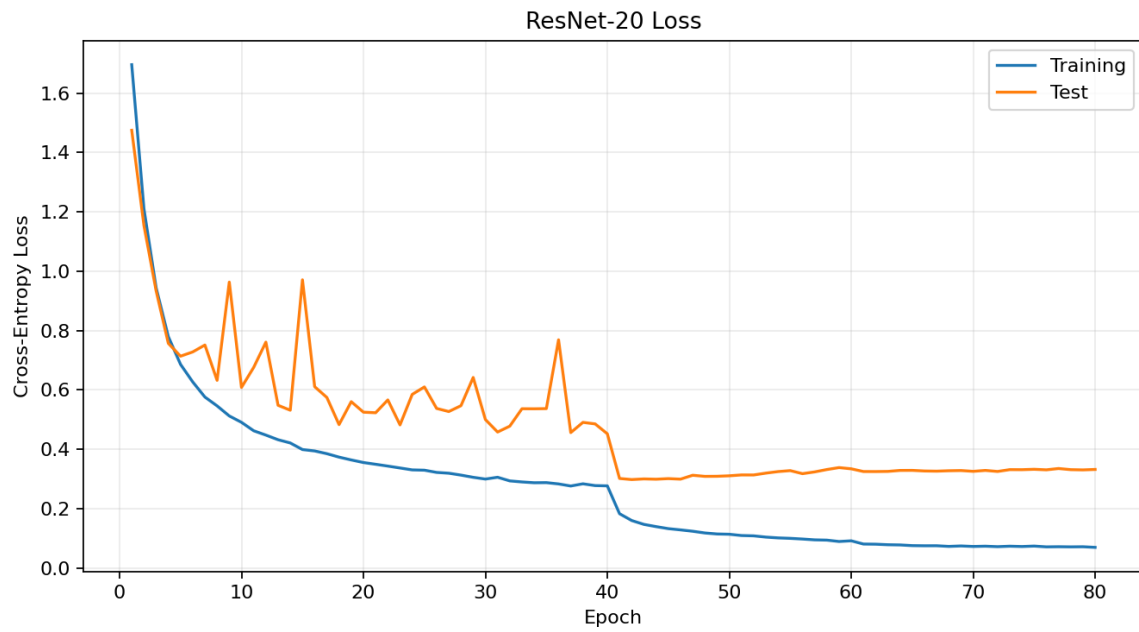
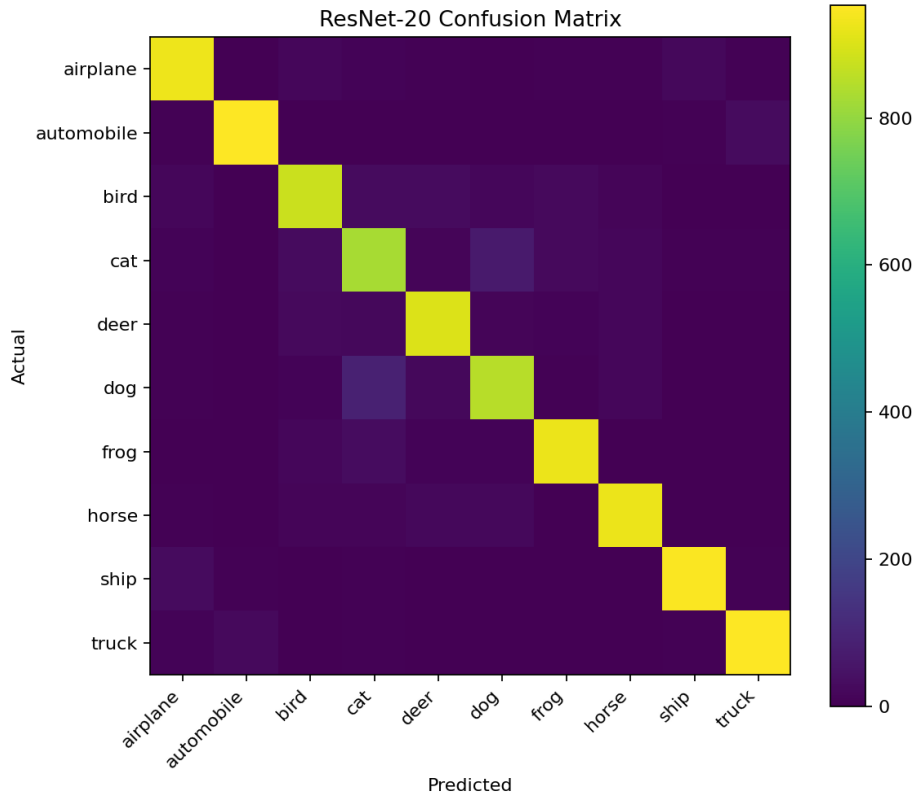


Figure 2. Loss curve.

9. Class-wise Evaluation

Class	Precision	Recall	F1
airplane	0.9162	0.9290	0.9225
automobile	0.9646	0.9530	0.9588
bird	0.8856	0.8750	0.8803
cat	0.8027	0.8300	0.8161
deer	0.9004	0.9040	0.9022
dog	0.8700	0.8500	0.8599
frog	0.9259	0.9250	0.9255
horse	0.9269	0.9260	0.9265
ship	0.9546	0.9470	0.9508
truck	0.9483	0.9530	0.9506

10. Confusion Matrix Analysis



The strongest observed confusions are dog→cat (88), cat→dog (68), frog→cat (32), ship→airplane (29), automobile→truck (28), and bird→cat / bird→deer (27 each).

11. Comparison With the Research Paper

Model	Parameters	Paper error	Current result
ResNet-20	0.27M	8.75%	9.08%
ResNet-32	0.46M	7.51%	Not yet
ResNet-44	0.66M	7.17%	Not yet
ResNet-56	0.85M	6.97%	Not yet

The current ResNet-20 error is 0.33 percentage points higher than the paper's reported 8.75% error. This is not a strict reproduction because the current schedule, hardware and evaluation procedure differ.

12. Methodological Limitation

The current notebook evaluates the test set after every epoch and selects the checkpoint using test accuracy. This introduces test-set leakage into model selection. Therefore, 90.92% is a prototype result, not a final unbiased estimate.

The paper uses a 45k/5k train/validation split to determine the training termination. The next version should use 45,000 training images, 5,000 validation images and reserve the 10,000 test images for the final evaluation.

13. Planned Final Experiments

1. Correct the train/validation/test split.
2. Retrain ResNet-20 using validation-based checkpoint selection.
3. Implement ResNet-32, ResNet-44 and ResNet-56.
4. Keep training conditions controlled so depth is the main variable.
5. Compare accuracy, error, parameter count and learning curves.
6. Add one controlled ablation or improvement after the baseline is stable.

14. Conclusion

A working research-paper-based ResNet-20 pipeline has been implemented on CIFAR-10. The current best observed accuracy is 90.92%. The immediate priority is methodological cleanup with a validation split, followed by the ResNet-20/32/44/56 depth comparison.

15. References

- [1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," supplied research paper.
- [2] CIFAR-10 dataset used in the project.
- [3] Project implementation notebook: P02.ipynb.